

Musterlösung · ML-Klausurtraining

Punkte in eckigen Klammern = Bewertungsschlüssel wie in der Klausur. Alle Schätzwerte sind nachgerechnet. Folgefehler werden berücksichtigt.

Aufgabe 1 (11 P)

- (a) $L(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum x_i}$ [2] · $l(\lambda) = n \log \lambda - \lambda \sum x_i$ [2]
(b) $l'(\lambda) = \frac{n}{\lambda} - \sum x_i$ [1] · $0 \Rightarrow \hat{\lambda} = \frac{n}{\sum x_i} = \frac{1}{\bar{x}}$ [2]
(c) $l''(\lambda) = -\frac{n}{\lambda^2} < 0$, da $n > 0$, $\lambda^2 > 0 \Rightarrow$ Maximum [2]
(d) $\hat{\lambda} = \frac{3}{6} = 0.5$ [1]
(e) $P(X > 2) \approx e^{-0.5 \cdot 2} = e^{-1} = 0.368$ [1]

Aufgabe 2 (11 P)

- (a) $L(\lambda) = \prod \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} = \frac{\lambda^{\sum x_i} e^{-n\lambda}}{\prod x_i!}$ [2] · $l(\lambda) = \sum x_i \log \lambda - n\lambda - \sum \log(x_i!)$ [2]
(b) $l'(\lambda) = \frac{\sum x_i}{\lambda} - n$ [1] · $0 \Rightarrow \hat{\lambda} = \frac{\sum x_i}{n} = \bar{x}$ [2]
(c) $l''(\lambda) = -\frac{\sum x_i}{\lambda^2} < 0$ (da $x_i \geq 0$, nicht alle 0) [2]
(d) $\hat{\lambda} = \frac{8}{4} = 2$ [1]
(e) $P(X = 0) = e^{-2} = 0.135$ [1]

Aufgabe 3 (10 P)

- (a) $L(\lambda) = \prod \frac{1}{2} \lambda^3 x_i^2 e^{-\lambda x_i} = 2^{-n} \lambda^{3n} (\prod x_i^2) e^{-\lambda \sum x_i}$ [2] · $l(\lambda) = -n \log 2 + 3n \log \lambda + 2 \sum \log x_i - \lambda \sum x_i$ [2]
(b) $l'(\lambda) = \frac{3n}{\lambda} - \sum x_i$ [1] · $\hat{\lambda} = \frac{3n}{\sum x_i} = \frac{3}{\bar{x}}$ [2]
(c) $l''(\lambda) = -\frac{3n}{\lambda^2} < 0$ [2]
(d) $\hat{\lambda} = \frac{6}{6} = 1$ [1]

Aufgabe 4 (10 P)

- (a) $L(\theta) = \theta^n (\prod x_i)^{\theta-1}$ [2] · $l(\theta) = n \log \theta + (\theta - 1) \sum \log x_i$ [2]
(b) $l'(\theta) = \frac{n}{\theta} + \sum \log x_i$ [1] · $\hat{\theta} = -\frac{n}{\sum \log x_i}$ [2]
(c) $l''(\theta) = -\frac{n}{\theta^2} < 0$ [2]
(d) $\sum \log x_i = 2 \log 0.5 = -1.3863 \Rightarrow \hat{\theta} = \frac{2}{1.3863} = 1.443$ [1] (Taschenrechner: ln!)

Aufgabe 5 (10 P)

- (a) $L(\theta) = 2^n \theta^n (\prod x_i) e^{-\theta \sum x_i^2}$ [2] · $l(\theta) = n \log 2 + n \log \theta + \sum \log x_i - \theta \sum x_i^2$ [2]
(b) $l'(\theta) = \frac{n}{\theta} - \sum x_i^2$ [1] · $\hat{\theta} = \frac{n}{\sum x_i^2}$ [2]
(c) $l''(\theta) = -\frac{n}{\theta^2} < 0$ [2]
(d) $\sum x_i^2 = 0.25 + 1 + 2.25 = 3.5 \Rightarrow \hat{\theta} = \frac{3}{3.5} = 0.857$ [1]

Aufgabe 6 (10 P)

- (a) $L(\beta) = \beta^n (\prod x_i)^{-(\beta+1)}$ [2] · $l(\beta) = n \log \beta - (\beta + 1) \sum \log x_i$ [2]
(b) $l'(\beta) = \frac{n}{\beta} - \sum \log x_i$ [1] · $\hat{\beta} = \frac{n}{\sum \log x_i}$ [2]
(c) $l''(\beta) = -\frac{n}{\beta^2} < 0$ [2]
(d) $\hat{\beta} = \frac{2}{2 \log 3} = \frac{1}{1.0986} = 0.910$ [1]

Aufgabe 7 (11 P)

- (a) $L(\theta) = \theta^{-n} e^{-\sum x_i/\theta}$ [2] · $l(\theta) = -n \log \theta - \frac{\sum x_i}{\theta}$ [2]
(b) $l'(\theta) = -\frac{n}{\theta} + \frac{\sum x_i}{\theta^2}$ [1] · $\cdot \theta^2 : -n\theta + \sum x_i = 0 \Rightarrow \hat{\theta} = \bar{x}$ [2]

$$(c) l''(\theta) = \frac{n}{\theta^2} - \frac{2\sum x_i}{\theta^3}; \text{ an } \hat{\theta} = \frac{\sum x_i}{n} \cdot \frac{n}{\hat{\theta}^2} - \frac{2n}{\hat{\theta}^2} = -\frac{n}{\hat{\theta}^2} < 0 \quad [2]$$

$$(d) \hat{\theta} = \frac{15}{3} = 5 \quad [1]$$

(e) θ ist die erwartete Lebensdauer; geschätzt hält ein Akku im Mittel 5 Jahre [1]

Aufgabe 8 (10 P)

$$(a) L(\theta) = \theta^{-n} \left(\prod x_i \right) e^{-\sum x_i^2 / (2\theta)} \quad [2] \cdot l(\theta) = -n \log \theta + \sum \log x_i - \frac{\sum x_i^2}{2\theta} \quad [2]$$

$$(b) l'(\theta) = -\frac{n}{\theta} + \frac{\sum x_i^2}{2\theta^2} \quad [1] \cdot \hat{\theta} = \frac{\sum x_i^2}{2n} \quad [2]$$

$$(c) l''(\hat{\theta}) = \frac{n}{\hat{\theta}^2} - \frac{\sum x_i^2}{\hat{\theta}^3} = \frac{n}{\hat{\theta}^2} - \frac{2n}{\hat{\theta}^2} = -\frac{n}{\hat{\theta}^2} < 0 \quad [2]$$

$$(d) \hat{\theta} = \frac{1+4+9}{6} = \frac{7}{3} = 2.333 \quad [1]$$

Aufgabe 9 (11 P)

$$(a) f(x; \lambda) = 2\lambda(\lambda x) e^{-(\lambda x)^2} = 2\lambda^2 x e^{-\lambda^2 x^2} \quad [1]$$

$$(b) L(\lambda) = 2^n \lambda^{2n} \left(\prod x_i \right) e^{-\lambda^2 \sum x_i^2} \quad [2] \cdot l(\lambda) = n \log 2 + 2n \log \lambda + \sum \log x_i - \lambda^2 \sum x_i^2 \quad [2]$$

$$(c) l'(\lambda) = \frac{2n}{\lambda} - 2\lambda \sum x_i^2 \quad [1] \cdot = 0 \Rightarrow \lambda^2 = \frac{n}{\sum x_i^2} \Rightarrow \hat{\lambda} = \sqrt{\frac{n}{\sum x_i^2}} \quad [2]$$

$$(d) l''(\lambda) = -\frac{2n}{\lambda^2} - 2 \sum x_i^2 < 0 \quad [2]$$

$$(e) \hat{\lambda} = \sqrt{2/2} = 1 \quad [1]$$

Aufgabe 10 (10 P)

$$(a) L(p) = p^n (1-p)^{\sum x_i} \quad [2] \cdot l(p) = n \log p + \sum x_i \log(1-p) \quad [2]$$

$$(b) l'(p) = \frac{n}{p} - \frac{\sum x_i}{1-p} \quad [1] \cdot n(1-p) = p \sum x_i \Rightarrow \hat{p} = \frac{n}{n + \sum x_i} = \frac{1}{1+\bar{x}} \quad [2]$$

$$(c) l''(p) = -\frac{n}{p^2} - \frac{\sum x_i}{(1-p)^2} < 0 \quad [2]$$

$$(d) \hat{p} = \frac{4}{4+4} = 0.5 \quad [1]$$

Aufgabe 11 (10 P)

$$(a) L(v) = (2\pi v)^{-n/2} e^{-\sum x_i^2 / (2v)} \quad [2] \cdot l(v) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log v - \frac{\sum x_i^2}{2v} \quad [2]$$

$$(b) l'(v) = -\frac{n}{2v} + \frac{\sum x_i^2}{2v^2} \quad [1] \cdot \hat{v} = \frac{\sum x_i^2}{n} \quad [2]$$

$$(c) l''(\hat{v}) = \frac{n}{2\hat{v}^2} - \frac{\sum x_i^2}{\hat{v}^3} = \frac{n}{2\hat{v}^2} - \frac{n}{\hat{v}^2} = -\frac{n}{2\hat{v}^2} < 0 \quad [2]$$

$$(d) \hat{\sigma}^2 = \frac{1+4+4+1}{4} = 2.5 \quad [1]$$

Aufgabe 12 (10 P)

$$(a) L(\lambda) = 6^{-n} \lambda^{4n} \left(\prod x_i^3 \right) e^{-\lambda \sum x_i} \quad [2] \cdot l(\lambda) = -n \log 6 + 4n \log \lambda + 3 \sum \log x_i - \lambda \sum x_i \quad [2]$$

$$(b) l'(\lambda) = \frac{4n}{\lambda} - \sum x_i \quad [1] \cdot \hat{\lambda} = \frac{4n}{\sum x_i} = \frac{4}{\bar{x}} \quad [2]$$

$$(c) l''(\lambda) = -\frac{4n}{\lambda^2} < 0 \quad [2]$$

$$(d) \hat{\lambda} = \frac{8}{8} = 1 \quad [1]$$

Aufgabe 13 (5 P)

$$(a) L(\theta) = \prod \frac{1}{\theta} = \theta^{-n}, \text{ falls } \theta \geq \max x_i, \text{ sonst } L(\theta) = 0 \quad [2]$$

(b) $L(\theta) = \theta^{-n}$ ist streng fallend, $l'(\theta) = -\frac{n}{\theta} \neq 0$ hat keine Nullstelle; L wird maximal für das kleinste zulässige θ : $\hat{\theta} = \max(x_1, \dots, x_n)$ [2]

$$(c) \hat{\theta} = 7 \quad [1]$$